

pst-art-led: PostScript LED Display

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26 July 2026

Contents

1 Metadata

Package File	pst-art-led.sty
Package Name	pst-art-led
Version Number	1.0.0
Release Date	2026/06/26
Dependencies	xcolor, pstricks
Author Name	Amit Manohar Manthanwar
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Repository	github.com/manthanwar/pst-art

2 Introduction

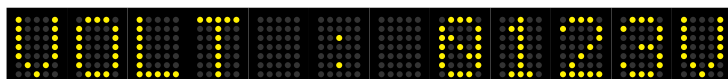
An LED 5×7 dot matrix is a rectangular grid of 35 light-emitting diodes arranged in 5 columns and 7 rows. It acts as a single, highly customizable display module used to show alphanumeric characters, simple graphics, or symbols in electronic projects. Because they are bright, compact, and affordable, they are widely used in hobbyist DIY builds and industrial panels as information boards or displays.

This PostScript package handles the input string and maps character data, translating text strings into precise column and row signals to lit the led using simple `ledColorOn=<color-name>` option passed to the `\psLed<csname>[<options>]{<string>}` command. It bridges the gap between retro aesthetic design and professional document typesetting. It allows users to programmatically generate scalable, clean matrix-style graphics, such as custom alphanumeric characters or pixel art, without relying on external image editors. It integrates pixel or matrix-style graphics natively into standard documents, making it ideal for technical documentation, science tutorials, or retro-computing projects.

3 String to LED Display

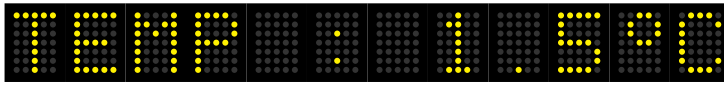
3.1 Example 1

```
\psset{unit=1mm,fillstyle=solid,linestyle=none}  
\psset{ledColorOn=yellow,ledWidth=8,ledDistance=12}  
\rput(10mm,-8mm){\psLedDisplay{Volt}}
```



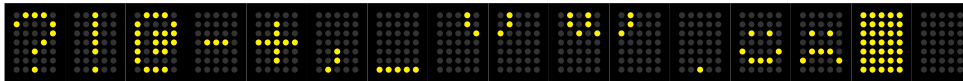
3.2 Example 2

```
\rput(0,-8mm){\psLedDisplay{Temp ~ : ~ 1.5 \degree C}}
```

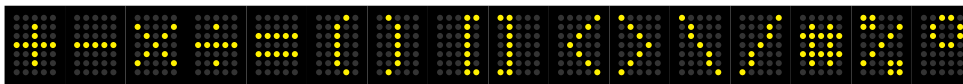


3.3 Symbols

```
\rput(0,-8mm){\psLedDisplay{? ! @ - + , _ .  
\quoteL \quoteR \quoteLD \quoteRD \smiley \smileySad \on \off}}
```



```
\rput(0,-8mm){\psLedDisplay{\plus \minus \multiply \divide  
= ( ) [ ] < > \backslash / \# \% \degree \on \off}}
```



Notice two different ways to get the plus sign using: + and \plus. Also, either ~ or \off results in space or led turned off.

3.4 Arrows and Currencies

```
\rput(0,-8mm){\psLedDisplay{\arrowL \arrowR \arrowU \arrowD  
\arrowNE \arrowNW \arrowSE \arrowSW  
\inr \eur \usd \gbp \rub \yen \won}}
```



3.5 Single Character

This  character 'A' is lit by `\rput(-4mm,0){\psLedDisplay[ledWidth=4, ledColorOn=green]{A}}`. Notice the clever use of \rput to precisely position the led. Make sure that the command options are set to `\psset{unit=1mm, fillstyle=solid,linestyle=none, ledColorOn=yellow, ledWidth=3}` prior to calling the command. We recommend you set these options globally. Note that the `ledWidth` is given in mm. The default led size is 20 mm and unit set to 1 mm. We recommend you use mm unit or convert other units for appropriate scaling before using.

4 Commands

The settings `\psset{<options>}` followed by command `\psLedDisplay{<string>}` or combined together `\psLedDisplay[<options>]{<string>}` achieves required results. This `\psLedDisplay` command internally uses the following set of characters.

4.1 Command options

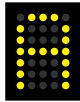
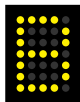
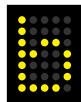
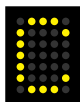
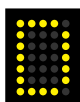
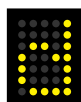
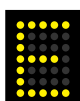
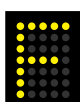
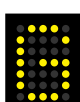
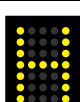
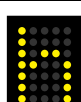
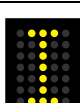
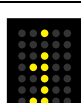
The following options control the behaviour of the script. They can be set by using `\psset{key=value}`. They can also be grouped together with comma separated key value pairs, i.e. a single comma followed

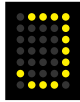
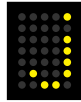
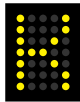
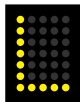
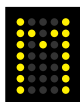
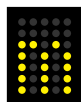
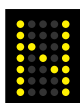
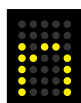
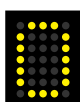
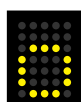
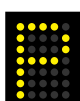
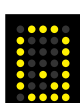
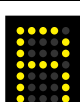
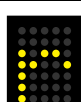
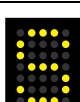
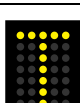
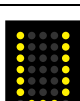
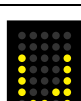
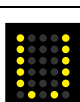
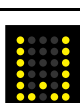
by multiple ‘option=value’ pairs.

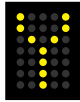
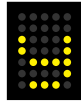
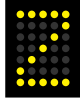
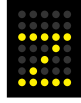
Key	Value	Remark
ledColorOn	color	<code>\psset{ledColorOn=red}</code>
ledColorOff	color	
ledColorBacklit	color	background color
ledWidth	color	
ledDistance	color	width + separation

For the choices of color names refer to [xcolor](#) package that provides driver-independent access to advanced color definitions, tinting, and blending.

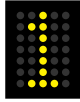
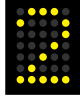
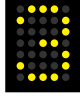
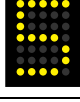
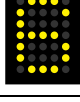
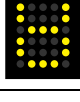
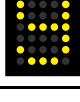
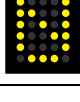
4.2 Letters

Command	Uppercase	Lowercase
<code>\psLeda \psLeda</code>		
<code>\psLedB \psLedb</code>		
<code>\psLedC \psLedc</code>		
<code>\psLedD \psLedd</code>		
<code>\psLedE \psLede</code>		
<code>\psLedF \psLedf</code>		
<code>\psLedG \psLedg</code>		
<code>\psLedH \psLedh</code>		
<code>\psLedI \psLedi</code>		

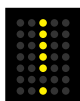
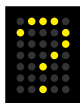
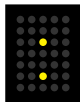
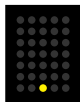
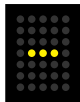
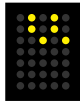
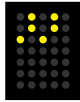
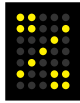
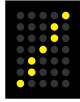
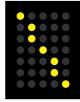
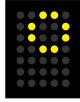
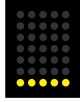
Command	Uppercase	Lowercase
<code>\psLedJ \psLedj</code>		
<code>\psLedK \psLedk</code>		
<code>\psLedL \psLedl</code>		
<code>\psLedM \psLedm</code>		
<code>\psLedN \psLedn</code>		
<code>\psLedO \psLedo</code>		
<code>\psLedP \psLedp</code>		
<code>\psLedQ \psLedq</code>		
<code>\psLedR \psLedr</code>		
<code>\psLedS \psLedS</code>		
<code>\psLedT \psLedt</code>		
<code>\psLedU \psLedu</code>		
<code>\psLedV \psLedv</code>		
<code>\psLedW \psLedw</code>		
<code>\psLedX \psLedx</code>		

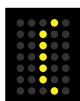
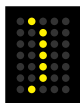
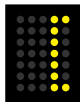
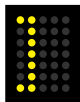
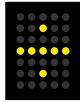
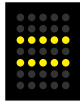
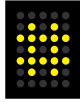
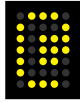
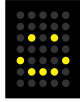
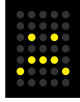
Command	Uppercase	Lowercase
<code>\psLedy</code> <code>\psLedy</code>		
<code>\psLedZ</code> <code>\psLedz</code>		

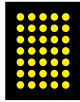
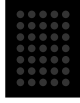
4.3 Numbers

Command	Output	Remark
<code>\psLedNA</code>		
<code>\psLedNB</code>		
<code>\psLedNC</code>		
<code>\psLedND</code>		
<code>\psLedNE</code>		
<code>\psLedNF</code>		
<code>\psLedNG</code>		
<code>\psLedNH</code>		
<code>\psLedNI</code>		
<code>\psLedNJ</code>		

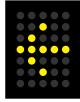
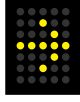
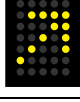
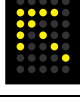
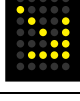
4.4 Symbols

Command	Output	Remark
<code>\psLedExclamation</code>		
<code>\psLedQuestion</code>		
<code>\psLedColon</code>		
<code>\psLedPeriod</code>		
<code>\psLedDash</code>		
<code>\psLedComma</code>		
<code>\psLedQuoteL</code>		single quote left
<code>\psLedQuoteR</code>		single quote right
<code>\psLedQuoteLD</code>		double quote left
<code>\psLedQuoteRD</code>		double quote right
<code>\psLedPercent</code>		
<code>\psLedSlash</code>		
<code>\psLedBackslash</code>		
<code>\psLedDegree</code>		
<code>\psLedUnderscore</code>		

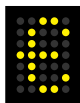
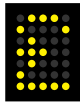
Command	Output	Remark
<code>\psLedBracketLeft</code>		
<code>\psLedBracketRight</code>		
<code>\psLedBracketSquareLeft</code>		
<code>\psLedBracketSquareRight</code>		
<code>\psLedBracketAngleLeft</code>		
<code>\psLedBracketAngleRight</code>		
<code>\psLedPlus</code>		
<code>\psLedMinus</code>		
<code>\psLedMultiply</code>		
<code>\psLedDivide</code>		
<code>\psLedEqual</code>		
<code>\psLedHash</code>		
<code>\psLedAmpersand</code>		
<code>\psLedSmiley</code>		
<code>\psLedSmileySad</code>		

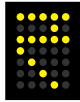
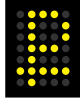
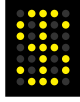
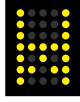
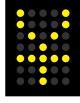
Command	Output	Remark
<code>\psLedOn</code>		
<code>\psLedOff</code>		

4.5 Arrows

Command	Output	Remark
<code>\psLedArrowL</code>		left
<code>\psLedArrowR</code>		right
<code>\psLedArrowU</code>		up
<code>\psLedArrowD</code>		down
<code>\psLedArrowNE</code>		north east
<code>\psLedArrowNW</code>		north west
<code>\psLedArrowSE</code>		south east
<code>\psLedArrowSW</code>		south west

4.6 Currency

Command	Output	Remark
<code>\psLedEUR</code>		Euro
<code>\psLedGBP</code>		Pound

Command	Output	Remark
<code>\psLedINR</code>		Rupee
<code>\psLedRUB</code>		Rubble
<code>\psLedUSD</code>		Dollar
<code>\psLedWON</code>		Won
<code>\psLedYEN</code>		Yen/Yuan

